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Final report of the course Advanced Machine Learning and Applications (IT2043.26.1.CH.01), Master of Information Technology, November 2026. Source code: github.com/haiiyins/retail_multidim_pipeline

Digital Object Identifier

Evaluating and Improving Personalized Product Recommendation with Multi-Signal Machine Learning on Large-Scale Retail Data

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This work builds on the data pipeline of the course Data Processing Engineering (IT2047.26.1.CH.01) of the same programme.

ABSTRACT Recommender evaluations of retail next-basket prediction often score a ranker only on the users whose candidate list already holds a purchase. On a loyalty data set of a multi-country sporting-goods retailer, this conditional protocol reported a normalized discounted cumulative gain at 20 of 0.206 for a gradient-boosted ranker over 200 rule-based candidates. Counting every purchase, including those the candidate generator never proposed, the same model scores 0.095, and a positive-level error decomposition shows that 58 % of the missed purchases were never retrieved, almost all of them items the member had not bought before. We therefore treat retrieval as a model: eight learned candidate sources, including linear auto-encoders, random-walk, nearest-neighbour, two-tower and self-attentive sequential models, are restricted to items new to the member, refitted every quarter in every split, tuned on the explore coverage they add, and merged by a rank-quota mixer under a cap of 300 candidates. Coverage rises from 0.370 to 0.450. Under one tuning budget, a tuned gradient-boosted ranker, a stacked variant and a deep multi-task ranker with repeat and explore heads and a learned gate are compared with five next-basket baselines. All two-stage rankers beat the strongest baseline by more than a quarter; the deep model gains about one percent over the tuned trees only in its single-head form, and the learned gate, not the explicit split, carries the benefit. Restricted to features a purchase log yields, the pipeline keeps about four fifths of its gain over the frequency rules a retailer uses today; the other data dimensions add the rest, mostly on new items. A controlled study confirms that naive stacking inflates in-sample metrics.

INDEX TERMS Evaluation methodology, gradient boosting, learning to rank, multi-task learning, next-basket recommendation, recommender systems, retail analytics, two-stage retrieval.

I. INTRODUCTION

A loyalty member of a multi-country sporting-goods retailer opens the app on a Monday: which product models will they buy in the next 30 days? A previous project built a point-in-time feature store over eleven exported tables and a LightGBM LambdaRank ranker [1], [2] (P1) over 200 candidates per (member, Monday) anchor. It reported a normalized discounted cumulative gain NDCG@20 of 0.206, but only on anchors whose candidate list contained at least one purchase. Measured over every anchor with a purchase, counting purchases outside the list as misses, the same model scores 0.090 (TH + VN: 0.095), and 40.7 % of the anchors have no purchase in the list at all.

This paper asks where those errors are and what it takes to fix them. It reformulates next-basket recommendation (NBR) as a two-stage problem in which retrieval is a model with its own objective, and compares a deep multi-task ranker against tuned gradient-boosted trees (GBDT) and strong NBR baselines under one tuning budget. The research questions are phrased as questions, not as expected orderings, because a GBDT win is a valid outcome [3]:

- **RQ1 (retrieval).** How much do learned candidate

sources raise unconditional recall over the rule-based sources R1–R5, separately for repeat and explore purchases, at what cost, and how much of that gain reaches NDCG@20 when the ranker is unchanged?

- **RQ2 (ranking).** With the same tuning budget and candidates, how does an end-to-end deep multi-task ranker (P3) compare with a tuned GBDT (P1-tuned), stacking (P2) and NBR baselines, overall and by segment?
- **RQ3 (repeat / explore).** Do explicit repeat and explore heads with a learned gate improve explore without hurting repeat, compared with a single head?
- **H2 (hypothesis).** Naive stacking inflates metrics on training and pseudo-test anchors relative to a rolling protocol; on the real test the gap narrows but does not vanish.
- **RQ4 (feasibility).** How many times better than "best-sellers" and "buy the same again" is the best system, and what purchase value does it capture (offline proxy), and how much of its gain needs data beyond the purchase history?

The contributions are: (1) an unconditional evaluation protocol with a repeat / explore split and a positive-level error decomposition that locates 58 % of the missed purchases in retrieval rather than ranking; (2) an explore-only retrieval layer of eight learned sources refitted quarterly in every split, with a rank-quota mixer; (3) a fair-budget comparison of P1-tuned, P2, P3 and five baselines on the same candidates; (4) a controlled measurement of stacking leakage (naive, rolling, leaky); and (5) a static, privacy-preserving demonstrator.

II. RELATED WORK

Next-basket recommendation. The reality check of Li et al. [4] showed that repeat purchases dominate common NBR metrics and that simple frequency baselines (P-TopFreq, GP-TopFreq) are hard to beat; it introduced the repeat / explore recall split used here. TIFU-KNN [5] models personal item frequency with time decay and neighbour smoothing. More generally, carefully tuned simple baselines often match recent neural recommenders [6].

Item-based and graph retrieval. EASE [7] is a closed-form linear auto-encoder; RP3beta [8] re-weights random walks by item popularity; item2vec [9] learns co-occurrence embeddings.

Neural retrieval and sequential models. Two-tower models [10] score users and items in a shared space with a softmax over items; SASRec [11] applies causal self-attention [12] to the behaviour sequence, gSASRec [13] corrects its overconfidence under negative sampling, and PinnerFormer [14] trains on a dense all-action objective over a future window. DIN [15] adds target attention of the candidate over the behaviour sequence.

Multi-task and repeat-aware ranking. MMoE [16] gates shared experts per task. Repeat-aware recommenders separate repeat and explore modes (RepeatNet [17], RE-CANet [18]). Our P3 gates two listwise heads with a learned member-level probability.

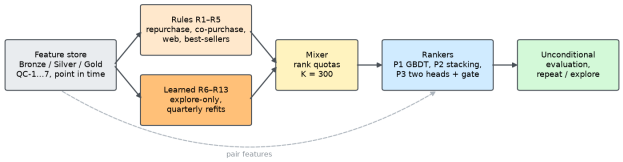


FIGURE 1. Two-stage system. Every learned component is fitted at quarterly cutoffs before the anchors it scores.

Leakage in stacking. Out-of-fold or temporally rolling fits are standard for stacked models [19]; we measure the inflation of an in-sample fit directly.

III. DATA AND INFRASTRUCTURE

The data set is an export of eleven tables from the retailer's data platform: point-of-sale and e-commerce lines (89.3 M lines), catalogue, price timeline, stock, loyalty members, favourite sports, stores, calendar, and 999.7 M web and app events, for Singapore (SG), Thailand (TH) and Vietnam (VN) from 2023-01 to 2026-10. A DuckDB / dbt lakehouse turns them into Bronze, Silver and Gold layers with seven quality gates (QC-1 to QC-7). After cleaning, the fact table holds 66.1 M sale lines in 26.3 M transactions by 4.7 M buying members. The unit of prediction is a (member, anchor Monday) pair; a positive is any non-returned model bought in $(t, t + \Delta]$ with $\Delta = 30$ days ($\Delta = 90$ as an auxiliary label).

Splits are by anchor date with a 42-day purge gap: train up to 2025-09-15, valid anchors 2025-10-27 to 2025-12-22, test anchors 2026-02-02 to 2026-07-27. Anchors are capped per country by hash rank (50 000 train, 15 000 valid, 50 000 test). The main scope is TH + VN; SG labels are reported but flagged REVIEW because of label quality. Unless stated otherwise, every number below is on the test split, TH + VN, $\Delta = 30$.

Every model-generated table carries its fit cutoff, the latest source timestamp it read and a model version, and is checked by QC-6: no row may read an event at or after its anchor, and no fit may use a cutoff later than the anchor. Learned candidate sources and signal models are refitted at 13 quarterly cutoffs (2023-07-01 to 2026-07-01) in every split; an anchor uses the latest cutoff $\leq t$. A single fit at the end of train would leave 27–34 % of the test positives outside a 5 000-model universe that is 5–10 months old; quarterly refits halve that share. Rankers are trained once on train and selected on valid. Fig. 1 shows the system.

The hardware is one shared GH200 node (72 Arm cores and one Hopper-class GPU); deep models use bfloat16 mixed precision. Every model with hyper-parameters receives the same budget of 10 Optuna [20] TPE [21] trials on valid. The course plan asked for 30 trials and three seeds; the rankers were re-tuned with 30 trials as a check (Section VII) and the final fits repeated with three seeds.

IV. UNCONDITIONAL EVALUATION AND ERROR ANALYSIS

TABLE 1. P1-frozen under the conditional and unconditional protocols (test, $\Delta = 30$).

Protocol, scope	anchors	NDCG@20	HR@10	rec. rep.	rec. exp.	cand.
conditional, TH + VN	25 538	0.206	0.368	0.826	0.218	1.000
unconditional, TH + VN	41 734	**0.095**	0.225	0.824	0.073	0.370
unconditional, all	63 574	0.090	0.217	0.797	0.065	0.351

TABLE 2. Fate of the test positives of P1-frozen over R1–R5 candidates (shares of positives).

	positives	top 10	rank 11+	capped	no source
repeat	18 251	0.654	0.332	0.013	0.002
explore	195 038	0.028	0.227	0.111	0.634
all	213 289	0.082	0.236	0.103	0.580

A. PROTOCOL

The evaluated unit is every used anchor a with at least one non-returned purchase in $(t, t + \Delta]$; its positives P_a are all of those purchases, including those outside the candidate list C_a, which are placed below every candidate. With L_a^K the top K of the ranked list,

$$\text{Recall@}K = \frac{|P_a \cap L_a^K|}{|P_a|}, \tag{1}$$

$$\text{NDCG@}K = \frac{\sum_{k=1}^{K} \mathbf{1}[\ell_k \in P_a] / \log_2(k+1)}{\sum_{k=1}^{\min(K,|P_a|)} 1/\log_2(k+1)}, \tag{2}$$

where ℓ_k is the item at rank k, so the ideal gain counts every positive, retrieved or not [22]. A positive is repeat if the member bought the model before t and explore otherwise [4]; recall_repeat@20 and recall_explore@20 average over the anchors that have positives of that kind. We also report HR@10 (at least one hit in the top 10) and the candidate recall (share of positives anywhere in C_a). Metrics are member means; confidence intervals come from a bootstrap over members [23] (1 000 resamples) and model differences from a paired bootstrap over the same members.

B. THE REFERENCE MODEL UNDER THE NEW PROTOCOL

Table 1 puts both protocols side by side. The conditional protocol drops the 40.7 % of the anchors whose list holds no purchase, which doubles the headline number.

C. WHERE THE ERRORS ARE

Each of the 213 289 test positives is put in one cell: ranked in the top 10, ranked below 10, proposed by a source but cut by the 200 cap, or proposed by no source (Table 2, Fig. 2).

Repeat is largely solved; explore is 91 % of the positives and 74.5 % of it never reaches the ranker. Of the capped positives, 95 % come from R5 (the national top 300), whose lower ranks do not fit the cap. The positives without a source are mid-tail, not long-tail: 95 % have a popularity rank between 301 and 5 000 at the anchor, so a learned source over a per-country universe of the top 5 000 models can reach them while losing only about 3 % of the positives. R4 (sport × offer, recall 0.0002) is dropped: only 1.07 % of the anchors have a favourite sport. These facts set the agenda of Section V.

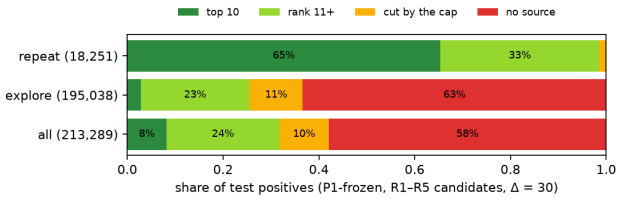


FIGURE 2. Where the test positives of the reference model end up.

TABLE 3. Learned candidate sources and their best valid gain in explore coverage over R1–R5@200 (10 trials).

Source	Model (tuned parameters)	Δ cand. exp.
R6	EASE, closed-form item–item (λ)	+0.085
R7	item2vec over baskets (defaults)	—
R8	TIFU-KNN (k, r_b, r_g, α)	+0.057
R9	two-tower, softmax over explore targets (d, τ, lr, dropout)	+0.074
R10	SASRec, dense all-action loss (d, blocks, dropout, lr)	+0.085
R11	explin, windowed linear (λ, half-life)	+0.074
R12	RP3beta, random walk (α, β)	+0.088
R13	context-aware trend (window, weights)	+0.069

V. LEARNED RETRIEVAL

A. SOURCES

R1 (repurchase) already holds 99.7 % of the repeat positives, so every learned source is explore-only: it masks the models the member bought before the anchor and is tuned on the explore coverage it adds to R1–R5@200 on valid (Δ cand_explore of the union), not on its own recall. Each source is fitted per country at each quarterly cutoff over the top 5 000 models by buyers in the previous 365 days, and proposes 100 candidates per anchor (Table 3). With 5 000 models per country, an exact matrix product on the GPU replaces an approximate nearest-neighbour index.

The two neural sources are adapted to the target. The two-tower model (R9) draws several reference days per member and epoch, adds the month of t to the member tower, and uses a softmax over the universe that masks the models already bought. SASRec (R10) takes one step per basket (the mean of its model embeddings) with a gap embedding, and is trained with a dense all-action explore loss in the spirit of PinnerFormer [14]: each step predicts every explore model of the following Δ days. R11 (explin) is a supervised windowed linear model $\hat{Y} = XB$ with closed-form ridge solution, where X is the time-decayed purchase history before t and Y the explore purchases in $(t, t + \Delta]$. R13 reweights the recent country popularity by the member's departments and price band.

TABLE 4. Candidate coverage on test (TH + VN, $\Delta = 30$).

Candidate set	list	cand.	repeat	explore
R1–R5 (reference)	200	0.370	0.997	0.307
R1–R5 ∪ R12 RP3beta (best single)	277	0.439	0.997	0.386
R1–R5 ∪ R6 / R10 / R7	260–282	0.430–0.435	0.997	0.375–0.381
R1–R5 ∪ all learned (uncapped)	483	0.541	0.997	0.499
candidates_v2, mixer, $K = 300$	300	**0.450**	0.972	**0.400**
mixer, $K = 500$	470	0.535	0.994	0.492

B. MIXER AND THE CAP

The ranker receives at most K candidates. The mixer interleaves the sources by weighted rank quotas: candidate i gets the key

$$\kappa(i) = \min_{s\,:\,i \in C_s} \frac{r_s(i) - 0.5}{w_s}, \tag{3}$$

where $r_s(i)$ is its rank in source s and w_s the source's stand-alone valid Recall@100, and the K smallest keys are kept. A per-member variant scales w_{R1} by the member's repeat share; it ties with the fixed policy on valid (TH + VN coverage at 300: 0.4841 against 0.4847), so the fixed policy is used. $K = 300$ is chosen from the recall–cost curve (Fig. 3).

C. COVERAGE

Coverage rises from 0.370 to 0.450 at $K = 300$ (Table 4), meeting the plan's target of 0.45; explore coverage rises 1.30 times (0.307 to 0.400), short of the target of 2. The cap costs 2.5 points of repeat coverage: R1 lists beyond rank 60 lose their slots to explore candidates. The sources are complementary: the union of all eight adds 0.17 over R1–R5 while the best single source adds 0.07. The neural sources do not beat the cheap ones: SASRec ties with EASE, the two-tower model is below, and RP3beta, the best single source, trains in minutes against about six GPU-hours of tuning and building for each neural source (Section VIII). QC-6 on candidates_v2 finds no row that reads an event at or after its anchor and no fit cutoff after the anchor. The pairs of candidates_v2 are rendered from the same dbt SQL as Gold and match it on a parity day up to floating-point noise.

D. FROM RECALL TO NDCG

With the ranker frozen (P1-frozen's hyper-parameters and original feature set, refitted on the v2 pairs), the retrieval gain reaches NDCG@20 only partly. The paired difference v2 − v1 is $+0.0025$ $[+0.0018, +0.0032]$ in NDCG@20 and $+0.0052$ in HR@10, with $+0.0101$ recall_explore@20 and -0.0315 recall_repeat@20. The gain is significant but small (0.0947 to 0.0973): the frozen ranker has no feature that tells it which learned source proposed a candidate, it trades repeat recall for explore recall, and most new explore candidates land below rank 20.

VI. RANKING MODELS

All rankers score the same candidates_v2 lists ($K = 300$) and are selected on unconditional valid NDCG@20 with 10 TPE trials.

- **P1-frozen**: the previous LambdaRank with its frozen hyper-parameters and original features, refitted on the v2 pairs; a reference row, not a tuned competitor.
- **P1-tuned**: LightGBM LambdaRank over every pair column (85 features, the source ranks of R6–R13 included).
- **P2**: P1-tuned plus five stacked signals fitted with the rolling protocol: a LightGBM department hazard over member × department rows, $P(\text{buy in } (t, t+\Delta])$, and LightGBM quantile models of willingness to pay (log unit price, quantiles 0.5 and 0.9).
- **P3 (UMTR)**: a Transformer over the last 50 purchases (model and log-gap embeddings), an MLP over the member context, an item tower over the candidate's pair features, and DIN-style target attention [15] of each candidate over the sequence states; 3.7 M parameters.
- **Baselines**: P-TopFreq, GP-TopFreq [4], EASE and TIFU-KNN as full-catalogue NBR rankers with their own 10-trial budget on valid NDCG@20, and SASRec (the R10 model without the explore mask, no separate budget).

P3 has two heads, repeat and explore, each with a softmax over its group of candidates (R_a: models bought before t; E_a: the others). A gate $g(u) \in (0, 1)$ computed from the member representation splits the probability mass:

$$p(i) = \begin{cases} g(u)\,\dfrac{e^{s_i^r}}{\sum_{j \in R_a} e^{s_j^r}}, & i \in R_a, \\[2ex] \big(1 - g(u)\big)\,\dfrac{e^{s_i^e}}{\sum_{j \in E_a} e^{s_j^e}}, & i \in E_a. \end{cases} \tag{4}$$

The loss is listwise, $\mathcal{L} = -\frac{1}{|P_a \cap C_a|} \sum_{i \in P_a \cap C_a} \log p(i) + \lambda_g \,\mathrm{BCE}(g, y^{rep}) + \lambda_{90}\,\mathcal{L}_{90}$, where y^{rep} is the repeat share of the member's positives and $\mathcal{L}_{90}$ an auxiliary binary cross-entropy on the $\Delta = 90$ label. Early stopping uses unconditional valid NDCG@20. The one-head ablation replaces the two group softmaxes by one softmax over C_a.

VII. RESULTS

A. RQ2: MODEL COMPARISON

Tables 5 and 6 and Fig. 4 are read as answers rather than as a leaderboard.

- Every two-stage ranker beats every NBR baseline by a wide margin (P1-tuned − TIFU-KNN: $+0.022$, 28 %). The strongest baseline is TIFU-KNN, then GP-TopFreq; EASE and SASRec as full-catalogue rankers fall below popularity because, without the repeat-preserving candidates of a two-stage system, they give up the repeats (recall_repeat 0.37–0.43 against 0.88).
- Tuning matters more than the model family: P1-tuned gains $+0.0039$ over the frozen parameters on the same pairs, more than the whole retrieval gain under the frozen ranker ($+0.0025$).
- The deep model and the GBDT are within a thousandth. P3 with one head is the only model significantly above P1-tuned ($+0.0009$, about 1 %), and only just above P2. The two-head P3 ties with P1-tuned: it keeps more

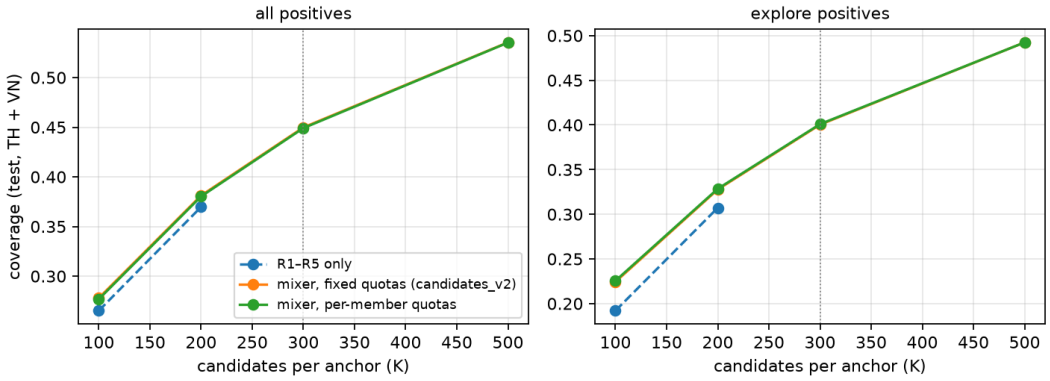


FIGURE 3. Candidate coverage against the cap K; the fixed and per-member mixers overlap at every K.

TABLE 5. Ranking on test (TH + VN, 41 734 anchors, $\Delta = 30$, unconditional), with 95 % member-bootstrap intervals of NDCG@20. Selection on valid with 10 trials and one seed.

Model	NDCG@20	95 % CI	HR@10	recall_repeat@20	recall_explore@20	cand. recall
P-TopFreq	0.0615	[0.0597, 0.0632]	0.156	0.877	0.000	0.093
GP-TopFreq	0.0752	[0.0734, 0.0769]	0.184	0.878	0.041	0.262
EASE NBR	0.0548	[0.0535, 0.0561]	0.146	0.368	0.072	0.231
TIFU-KNN NBR	0.0788	[0.0770, 0.0806]	0.187	0.885	0.043	0.234
SASRec NBR	0.0606	[0.0589, 0.0622]	0.156	0.435	0.067	0.235
P1-frozen (R1–R5 candidates)	0.0947	[0.0925, 0.0968]	0.225	0.824	0.073	0.370
P1-frozen (v2)	0.0973	[0.0949, 0.0994]	0.230	0.793	0.083	0.450
P1-tuned (v2)	0.1011	[0.0990, 0.1033]	0.239	0.817	0.092	0.450
P2 (v2)	0.1013	[0.0994, 0.1034]	0.240	0.806	0.093	0.450
P3 two heads	0.1009	[0.0988, 0.1029]	0.237	0.845	0.089	0.450
P3 one head	**0.1020**	[0.0999, 0.1040]	0.240	0.838	0.090	0.450

TABLE 6. Paired member-bootstrap differences on test (TH + VN), $a - b$ [95 % CI].

$a - b$	NDCG@20	rec. explore@20
P1-tuned − P1-frozen (v2)	+0.0039 [+0.0032, +0.0045]	+0.0085
P2 − P1-tuned	+0.0002 [−0.0002, +0.0006]	+0.0007
P3 two heads − P1-tuned	−0.0002 [−0.0009, +0.0005]	−0.0031
P3 one head − P1-tuned	+0.0009 [+0.0002, +0.0016]	−0.0016
P3 one head − P2	+0.0007 [+0.0000, +0.0014]	−0.0023
P1-tuned − TIFU-KNN NBR	+0.0224 [+0.0214, +0.0234]	+0.0488

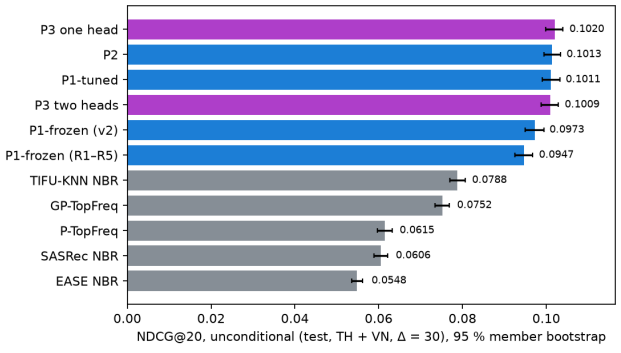


FIGURE 4. NDCG@20 of every model on test with 95 % member-bootstrap intervals.

repeats (+0.029 recall_repeat@20) and finds fewer explore purchases (−0.003).

- Stacking adds nothing measurable (P2 − P1-tuned +0.0002, interval across zero), consistent with Section VII-D: the Gold pair features already hold department recency, cycles and price gaps.

On valid, over the three countries of the tuning objective, the order is the same (P3 one head 0.1049, P3 two heads 0.1037, P2 0.1034, P1-tuned 0.1033).

Seed variance. The final fits of P1-tuned and both P3 variants were repeated with seeds 1 and 2 at the same hyperparameters (Table 7, Fig. 5). The seed spread ($\leq$ 0.0003) is below the gap between P3 one head and P1-tuned; the seed-matched paired differences are +0.0009, +0.0010 and +0.0006, and one head beats two heads at every seed (+0.0011, +0.0018, +0.0011). The two-head model is the least stable: its repeat recall moves from 0.79 to 0.85 across seeds, i.e., the learned gate lands in different places from one run to the next.

Is the cap binding? Cutting the mixer list at $K = 500$ instead of 300 raises test coverage from 0.450 to 0.535 for 1.6 times the pairs. P1-frozen was refitted as is, P1-tuned and P3 were re-tuned with 10 trials. On valid (TH + VN), the decision data, NDCG@20 does not improve: P1-tuned 0.1088 to 0.1079, P3 two heads 0.1087 to 0.1092,

TABLE 7. Seed variance of the final fits (test, TH + VN, NDCG@20).

Model	seeds 0 / 1 / 2	mean ± sd	rec. repeat mean ± sd
P1-tuned	0.1011 / 0.1013 / 0.1016	0.1013 ± 0.0002	0.809 ± 0.007
P3 two heads	0.1009 / 0.1005 / 0.1011	0.1008 ± 0.0003	0.824 ± 0.030
P3 one head	0.1020 / 0.1022 / 0.1022	0.1022 ± 0.0001	0.831 ± 0.008

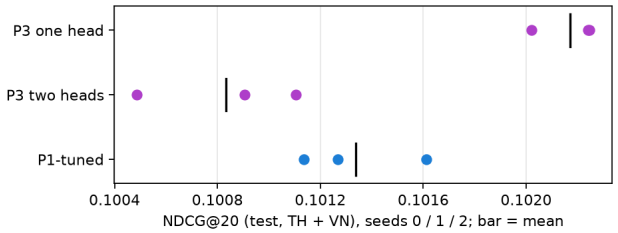


FIGURE 5. NDCG@20 of the final fits across three seeds.

P3 one head 0.1096 to 0.1097, so $K = 300$ is kept. Reported once afterwards on test (paired, $K = 500 - K = 300$): P1-frozen -0.0009 $[-0.0016, -0.0003]$, P1-tuned -0.0003 $[-0.0008, +0.0001]$, P3 two heads $+0.0006$ $[-0.0000, +0.0011]$, P3 one head $+0.0012$ $[+0.0007, +0.0017]$ (0.1032). The one-head test gain is not supported on valid and is not used for selection, which would select on test. The GBDT loses 0.025 repeat recall to the extra negatives. The cap is not what limits accuracy; ranking the hard explore candidates is.

Is the tuning budget binding? With 10 trials the best GBDT trial was the last one and sat near the learning-rate floor, and the best P3 trial sat at its floor. P1-tuned, P2 and P3 were therefore re-tuned on the same v2 pairs with 30 trials each and lower floors (GBDT learning rate ≥ 0.005 with up to 6 000 rounds, P3 learning rate $\geq 2 \cdot 10^{-5}$), the same budget for every ranker. On valid (TH + VN, the decision data), NDCG@20 does not improve from 10 to 30 trials: P1-tuned 0.1088 to 0.1089, P2 0.1089 to 0.1088, P3 two heads 0.1087 to 0.1086, P3 one head 0.1096 to 0.1092, so the 10-trial fits are kept. Reported once afterwards on test (paired, $30 - 10$ trials): P1-tuned $+0.0001$ $[-0.0003, +0.0006]$, P2 $+0.0007$ $[+0.0003, +0.0011]$ (0.1020), P3 two heads -0.0002 $[-0.0005, +0.0001]$, P3 one head -0.0000 $[-0.0003, +0.0003]$ (0.1020). The search had converged at 10 trials, and the order of Table 5 stands: with 30 trials P2 ties with P3 one head on test, but it does not on valid.

B. BY COUNTRY AND SEGMENT

Table 8 splits NDCG@20 by country and by member segment (transactions in the 365 days before the anchor: new ≤ 1, mid 2, loyal ≥ 3; all countries). The ordering holds in every country and segment, and the differences between P1-tuned and P3 stay below 0.002 everywhere. The largest gain of the whole pipeline over the reference is on new members (0.079 to 0.090, +14 %), whose purchases are almost all explore and who benefit most from the learned sources. SG (REVIEW scope) gains as much as TH and VN. A web / no-web split is not reported: 2–19 % of the anchors have web events and the web features carry about 1 % of the SHAP mass (Section VII-E).

TABLE 8. NDCG@20 on test by country and segment.

Model	SG	TH	VN	new	mid	loyal
TIFU-KNN NBR	0.069	0.085	0.072	0.065	0.080	0.078
P1-frozen (v1)	0.082	0.101	0.088	0.079	0.094	0.095
P1-tuned (v2)	0.089	0.108	0.093	0.089	0.101	0.099
P3 two heads	0.089	0.107	0.093	0.088	0.101	0.100
P3 one head	0.090	0.109	0.094	0.090	0.101	0.100

TABLE 9. P3 ablation, paired differences against P3 with two heads and a learned gate (test, TH + VN).

Variant − two heads	NDCG@20	HR@10	rec. rep.	rec. exp.
one head	+0.0011	+0.0025	−0.0073	+0.0016
fixed gate	−0.0066	−0.0105	−0.2753	+0.0159
no sequence	−0.0003	+0.0001	−0.0240	+0.0020

NDCG@20 intervals: one head [+0.0007, +0.0016]; fixed gate [−0.0074, −0.0058]; no sequence [−0.0007, +0.0000]. Fixed gate: g = the member's historical repeat share.

C. RQ3: REPEAT AND EXPLORE HEADS

The explicit split does not improve explore (Table 9, Fig. 6): the single head is better on NDCG@20 and on explore recall, at a small cost in repeat recall. The gate is the part that matters: replacing the learned g by the member's historical repeat share loses 0.0066 NDCG@20 and a third of the repeat recall. A plausible reading is that the historical share (about 10 % of positives) under-weights the repeat group for the anchors where a repeat is due, which the learned gate can see from the context. The learned gate ranks segments correctly (mean g: loyal 0.23, mid 0.17, new 0.14) but is not calibrated: it is about twice the observed repeat share (0.12, 0.08, 0.06) and correlates only weakly with the per-anchor target (0.11–0.19). The sequence encoder adds repeat recall ($+0.024$) but no NDCG: the pair features already summarize the history.

D. H2: STACKING LEAKAGE

The department hazard is fitted three ways: rolling (quarterly cutoffs, the only point-in-time-correct variant), naive (fitted once on train and applied in-sample) and leaky (fitted on every split). Its AUC on train / pseudo-test (last train quarter) / test is 0.681 / 0.703 / 0.708 for rolling, 0.725 / 0.744 / 0.705 for naive, and 0.710 / 0.717 / 0.723 for leaky. The in-sample fit looks 0.04 AUC better on the anchors it has seen and loses that lead on test; the leaky fit stays better on test because it has seen those labels.

Fused into a ranker (P1-frozen parameters, fitted on train anchors up to 2025-05-31), the signals give the NDCG@20 of Table 10. H2 holds in direction on both candidate sets, more clearly on v2: the naive fusion is ahead of rolling on its own training anchors ($+0.0026$) and on the pseudo-test ($+0.0029$), and behind it on test (-0.0004). The leaky fusion is the only one ahead on test ($+0.0011$ on v2), because its

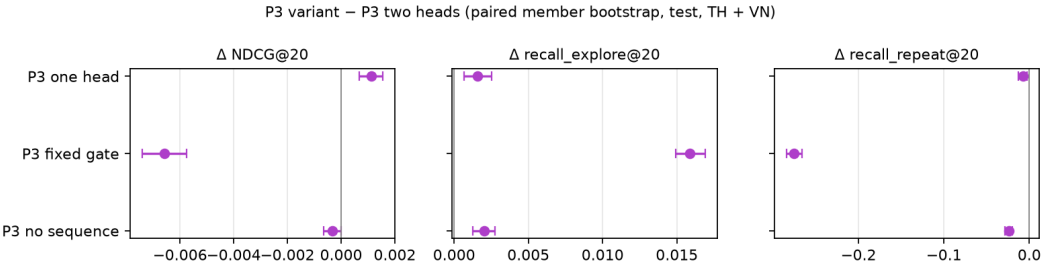


FIGURE 6. P3 ablation on test.

TABLE 10. Stacking leakage: unconditional NDCG@20 of a fused ranker by signal protocol (TH + VN).

	v1 candidates			v2 candidates		
Signals	train	pseudo	test	train	pseudo	test
none	0.1219	0.1072	0.0941	0.1246	0.1097	0.0961
rolling	0.1227	0.1090	0.0945	0.1255	0.1097	0.0964
naive	0.1246	0.1095	0.0934	0.1281	0.1126	0.0959
leaky	0.1232	0.1087	0.0944	0.1256	0.1091	0.0975

TABLE 11. P1-tuned: share of mean $|\mathrm{SHAP}|$ by feature group (test).

Feature group	all	repeat cand.	explore cand.
model: popularity / season / stock	0.475	0.298	0.492
retrieval: learned source ranks	0.165	0.202	0.162
retrieval: R1–R5 flags / scores / rank	0.085	0.140	0.079
member: price profile	0.080	0.031	0.085
pair: own history	0.027	0.155	0.014
web behaviour	0.013	0.013	0.013

signals were fitted on the test labels. The inflation is small because the signals are weak given the Gold features, and the gaps are of the order of the member-bootstrap intervals (± 0.002 on test, ± 0.005 on the 5 337 pseudo-test anchors). An in-sample stacked signal would nevertheless have been reported as a gain that does not exist on test.

E. WHAT THE GBDT USES

Grouped TreeSHAP [24] over 3 000 test anchors (Table 11) shows that the learned source ranks are the second feature group of P1-tuned (17 % of the mean absolute SHAP value), which is why the tuned rankers convert more of the retrieval gain than the frozen one, which does not see them. In P2 the five stacked signals take 2.4 % of the mass.

VIII. FEASIBILITY AND DEMONSTRATOR

A. VALUE OVER SIMPLE RULES

The best model on valid, P3 with one head, is compared with three references on test HR@10 (Table 12). "Random" draws 10 slots uniformly from the 5 000-model universe (analytic); "popular" is the country top 10 by buyers in the 28 days before the anchor, the same list for every member; "buy-again" is P-TopFreq. The model is 1.8 times the best-sellers

TABLE 12. HR@10 of P3 one head against business references (test, TH + VN).

Positives	P3	random	popular	buy-again	× pop.	× again
all	0.240	0.007	0.137	0.156	1.76	1.54
repeat	0.775	0.003	0.178	0.777	4.36	1.00
explore	0.110	0.006	0.115	0.000	0.96	—

and 1.5 times "buy the same again" overall. On repeats it matches buy-again (both about 0.78); on explore purchases it is slightly below the best-sellers (0.110 against 0.115). Its value is that it does both at once in one list of ten, which no single rule does.

As an offline proxy, not a causal estimate, the GMV-weighted Recall@10 is 0.077 against 0.053 for buy-again and 0.023 for popular, i.e., 268 EUR of observed next-30-day purchase value per 1 000 recommendations against 184 and 79. These purchases happened anyway; an online test is needed to know how many a recommendation would cause.

Purchase history only. The status quo recommends from purchase history alone. To separate what the extra data dimensions bring from what the two-stage pipeline brings, P1 was re-tuned (10 trials) and refitted on the same pairs with only the 39 of 85 features a purchase log yields: R1, R2, R5 and the learned source ranks, own and department history, co-purchase, sales popularity, season and model age, activity counts and country. Web behaviour (R3 and the mixed candidate rank), prices and promotions, stock, stores, calendar, sport affinity and the member price profile are removed. Table 13 crosses data and method. The pipeline on history alone gains $+0.0214$ $[+0.0204, +0.0225]$ NDCG@20 over GP-TopFreq; the other dimensions add $+0.0045$ $[+0.0039, +0.0052]$ for P1-tuned and $+0.0054$ $[+0.0046, +0.0062]$ for P3 one head over P1-hist (paired, test; valid gives the same order: 0.0819, 0.1044, 0.1088, 0.1096). Of the $+0.0268$ of P3 one head over GP-TopFreq, about four fifths are therefore reached by retrieval and a learned ranker on the purchase log, and one fifth needs the other data, mostly as explore recall (P1-tuned − P1-hist $+0.0077$, repeat -0.0086). The history-only row is slightly optimistic: its candidate pool is the multi-dimensional one (it still holds the web-intent candidates, without the features

TABLE 13. Data × method on test (TH + VN, NDCG@20). P1-hist: P1 re-tuned on the 39 purchase-history features.

Data	Method	Model	NDCG@20
purchase history	frequency rule	GP-TopFreq	0.0752
	NBR	TIFU-KNN	0.0788
	pipeline	P1-hist	0.0966
multi-dimensional	pipeline	P1-tuned	0.1011
		P3 one head	**0.1020**

TABLE 14. Cost of the candidate sources (13 cutoffs × 3 countries).

Source	device	tune h	build h	Δ exp.	per build h
R6 EASE	GPU	0.04	0.07	+0.085	1.27
R12 RP3beta	GPU	0.05	0.08	+0.088	1.16
R8 TIFU-KNN	GPU	0.04	0.08	+0.057	0.75
R13 trend	CPU	0.03	0.13	+0.069	0.54
R11 explin	GPU	0.07	0.47	+0.074	0.16
R7 item2vec	CPU	—	0.56	—	—
R9 two-tower	GPU	2.48	3.05	+0.074	0.02
R10 SASRec	GPU	4.19	3.18	+0.085	0.03

that explain them), and the two-tower and trend sources use the unit price paid on the sales lines.

B. SERVING COVERAGE

Every evaluated test anchor gets a personalized list of 300 candidates (median 300; all anchors have at least 10). This holds by construction, because the anchor grid starts after a member's second purchase; members without a purchase history are outside the evaluated population and get the country best-sellers at serving time.

C. COST

Table 14 lists the cost of the candidate sources on the shared GH200 (wall times are upper bounds). The neural sources cost 50–100 times more compute than EASE or RP3beta for the same or a lower coverage gain. Among the rankers the costs are all small at this scale (150 000 training anchors × 300 candidates): tuning takes 0.22–0.38 h, a final fit 1.3–5.1 min, and P3 (3.7 M parameters) scores 1 000 test anchors in 0.09 s on the GPU.

D. DEMONSTRATOR

A static page on a public Hugging Face Space runs the recommender in the browser (Pyodide). The user uploads a purchase file (member, country, model, purchase date, quantity); the page validates it and returns the top K of each member with the source of each item (repeat with time decay, co-purchase from EASE neighbours, recent country trend), merged with the same rank-quota rule as the mixer, and a reason in words. Model identifiers are salted hashes, departments carry generic labels, popularity is shown as a percentile, the sample members are synthetic, and the salt never leaves the server. The demonstrator uses reference tables frozen at 2026-07-28 and does not run the trained rankers: it shows the retrieval layer and its explanations, not the accuracy of Table 5.

IX. DISCUSSION AND LIMITATIONS

GBDT against deep models. Under the same budget and the same candidates, the deep multi-task ranker gains about 1 % NDCG@20 over a tuned GBDT and only in its single-head form, in line with the tabular-data literature [3]. The effort paid most in retrieval (+22 % relative coverage) and in tuning (+4 % NDCG@20 on the frozen ranker).

Method against data. Most of the gain over the status quo comes from the method, not from the extra data: on purchase history alone the pipeline reaches 0.0966 against 0.0752 for GP-TopFreq, and the web, price, promotion, stock and store features add 0.005 NDCG@20 on top. A retailer with only a sales log can therefore take most of the benefit; the other dimensions matter for the explore purchases.

Explore is still the bottleneck. Explore coverage rose 1.30 times, not 2, and at the top 10 the model is no better than the best-sellers on explore purchases. On average 60 % of a member's explore purchases are still outside the 300 candidates. Raising the cap to 500 recovers coverage (0.535) but not NDCG@20 on valid: the extra candidates are explore models the rankers cannot yet place in the top 20.

Tuning budget. The selected fits use 10 trials; a 30-trial re-tuning of every ranker changed valid NDCG@20 by at most 0.0004 and was not adopted. The final fits were repeated with three seeds: the seed spread ($\leq$ 0.0003 NDCG@20) is smaller than the P3 one-head gain over P1-tuned. The searches themselves ran with one sampler seed, and the candidate sources and NBR baselines kept the 10-trial budget; SASRec NBR has no NDCG@20 budget of its own. The margins between P1-tuned, P2 and P3 are below 0.001; P2's place depends on the budget (30 trials bring it level with P3 one head on test, not on valid), and only the gain of P3 one head over P1-tuned holds at every seed and at both budgets.

Offline only. GMV figures are a proxy; recommendations change behaviour, which an offline replay cannot measure.

Scope. SG labels are REVIEW; SG results are reported but not used for conclusions. gSASRec, distillation of P3 into a GBDT, conformal intervals, drift monitoring, and the what-if and parity check of the demonstrator were dropped from the plan.

Future work. A full-catalogue universe with monthly refits addresses the coverage ceiling of the 5 000-model universe; content embeddings of the catalogue text and the web click-stream as a second behaviour sequence, possibly with a generative sequential model such as HSTU [25], target the explore purchases that no co-purchase signal reaches.

X. CONCLUSION

Measured over every purchase, the reference recommender was half as good as reported, and most of its misses were purchases it never saw. Treating retrieval as a model, with eight explore-only sources refitted quarterly and mixed under a 300-candidate cap, raised the share of purchases within

reach from 0.370 to 0.450; with tuned rankers that see the source ranks, NDCG@20 rose from 0.095 to 0.1011–0.1020 (+7–8 %), more than 25 % ahead of the best next-basket baseline (0.0788). A deep two-head ranker with a learned gate does not beat a tuned GBDT; a single-head version does by 1 %. The learned gate is essential; the explicit split is not. On purchase history alone the pipeline keeps about four fifths of its gain over the frequency rules; the other data dimensions add the rest. Naive stacking shows the inflation H2 predicted, small because the signals add little over a good feature store.

ACKNOWLEDGMENT

This work was carried out as the project of the course Advanced Machine Learning and Applications (IT2043.26.1.CH.01) of the Master of Information Technology programme at the University of Information Technology, Vietnam National University Ho Chi Minh City, under the supervision of Dr. Ha Minh Tan, whom the authors thank for the guidance. The data set was made available for academic use by its owner; no organisation-specific identifier appears in this report or in the repository.

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